

MPLADS-Sentinel: AI-Powered Monitoring & Analytics Platform for MPLADS

Project Corpus — Problem Context, Research, Features, Architecture & Build Plan

1. Problem Context

1.1 The Scheme

The Members of Parliament Local Area Development Scheme (MPLADS) allows MPs to recommend developmental works in their constituencies, funded through an annual entitlement. The scheme lifecycle runs through five stages: **Recommendation → Sanction → Progress → Payment → Completion**, involving MPs, District Authorities, State Nodal Authorities, Implementing Agencies, and the Ministry of Statistics & Programme Implementation (MoSPI) as the Central Nodal Agency.

1.2 The Core Problem

High transaction volume + manual oversight across a five-stage, multi-actor pipeline creates blind spots:

- Irregularities, unusual expenditure patterns, and potential fund misuse go undetected
- Project delays, cost overruns, and duplicate work allocations across districts are hard to catch
- Non-compliance with scheme guidelines lacks real-time, cross-stakeholder visibility

1.3 The Data Source: eSAKSHI Portal

- **Live since 1 April 2023**, built by MoSPI in collaboration with SBI, replacing the earlier physical-mode process (manual documentation, district-level physical bank accounts)
- Digitizes the entire fund-flow: MP recommendation → District Authority sanction → Implementing Agency execution & payment requests → completion marking
- Captures **geotagged asset photos** and sanction documents at each stage

- **Critical data limitation:** historical data for the 17th Lok Sabha exists only from **2023–24 onward**; pre-2023–24 data (2019–20 through 2022–23) and all pre-2023–24 Rajya Sabha data is **not available** on the portal. This directly constrains what ML approaches are viable (see §4).
 - Data is updated in real time, but **entered manually by different stakeholders at each stage** — this multi-actor entry structure is itself a strong signal source (cross-stage consistency checks) rather than purely a limitation
 - Completion status is **self-reported by Implementing Agencies** — a known integrity gap (a work can be fully paid but never marked complete, or marked complete without verification)
 - Guideline timelines exist and are checkable: e.g., Para 3.2.4 mandates sanction/rejection within 45 days of recommendation receipt; Para 3.2.12 mandates a stipulated completion timeline in the sanction letter itself
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2. Research Findings: Real-World Grievances & Design Lessons

No direct public discussion of MPLADS/eSAKSHI exists on Reddit or forums (too niche a portal). Research was broadened to (a) how Indian citizens/officials discuss government portals generally, and (b) documented failure patterns in fraud-detection and alerting systems. Four patterns emerged, each mapped to a concrete design decision.

2.1 Pattern: “Indian government portals fail under load and erode trust”

Evidence: IRCTC session-timeout memes, Income Tax e-filing portal crashing for days post-launch (2021), Infosys being publicly summoned by the Finance Minister over unresolved glitches, driving-license portals citing “slow server” as a chronic excuse.

Design response:

- Horizontally scalable, partitioned ingestion (Kafka partitioned by state/district) instead of a monolithic write path — this exact bottleneck killed the IT e-filing portal at peak season; MPLADS has an analogous peak near fiscal year-end fund-utilization deadlines
- Offline-first mobile data entry with local caching and sync-on-reconnect, so field-level staff don’t lose work during connectivity drops
- A **meta-monitoring layer**: the platform tracks its own uptime/latency/error rates as a first-class dashboard, not an afterthought

2.2 Pattern: False positives are the primary failure mode of fraud-detection

systems

Evidence from fraud/AML literature: false positives are usually caused by poor data quality, overly strict matching algorithms, and insufficient contextual information; every wrongly flagged case carries a real cost that's often invisible because it's treated purely as a model-quality metric rather than an institutional cost.

Why this matters more for MPLADS than for banking fraud: a false positive here doesn't just waste an analyst's time — it can wrongly implicate an MP or District Authority, which is politically and legally consequential. Precision must be prioritized over recall for anything that becomes a public-facing flag.

Design response:

- **Two-tier alerting:** Tier 1 = silent internal risk-score adjustment (never shown to a human); Tier 2 = surfaced to investigators, and only promoted to Tier 2 after corroboration across **two or more independent signal types** (e.g., cost anomaly *and* photo/progress mismatch — not either alone)
- Every human-reviewed alert (confirmed or dismissed) feeds a monthly retraining loop — closing a feedback cycle that static, rule-based government systems typically never build

2.3 Pattern: Alert fatigue silently defeats monitoring systems

Evidence from SRE/data-engineering practice: when alert volume is high and much of it isn't actionable, teams stop trusting the alerting channel entirely and miss the alerts that actually mattered — the same dynamic documented in real incident post-mortems (e.g., desensitization to track-circuit alarms preceding a transit incident).

Design response:

- Role-scoped, volume-capped alerts: a Ministry official sees a curated weekly digest of nationally significant flags, not a raw firehose; District Authorities see only their jurisdiction
- Every alert carries a **recommended next action** (e.g., "request updated geotagged photo," "cross-verify vendor GSTIN"), never just a bare anomaly score
- Built-in alert-quality audit: track precision-per-alert-type over time as a first-class metric, surfaced to platform admins

2.4 Pattern: Digital divide — language, connectivity, and literacy at the point of data entry

Evidence: most Indian government digital interfaces are limited to English/Hindi, excluding

citizens and field officials literate only in regional languages; rural internet penetration and speeds remain well below urban levels, and BHASHINI (India's national AI language-translation platform) is designed to address this gap but is still early in rollout.

Why this matters: District/Implementing Agency staff entering sanction, progress, and payment data are frequently at block/panchayat level — not urban, not necessarily English/Hindi-first.

Design response:

- **BHASHINI integration** for data-entry forms and dashboards in regional languages — a genuinely current, India-specific, technically credible feature most competing proposals will overlook
- Low-bandwidth mode: compressed image uploads, progressive dashboard loading, SMS/IVR fallback for status alerts to field staff without reliable data access

3. Feature Set (Consolidated)

#	Feature	Primary User	What It Solves
1	Duplicate/overlapping work detection	District/State Authority	Same work sanctioned twice under different names across adjacent constituencies
2	Cost anomaly detection	District Authority, Ministry	Abnormal per-unit costs vs. category/region benchmarks
3	Delay & overrun early-warning	MP, District Authority	Predicts deviation from guideline timelines (45-day sanction rule, stipulated completion date) before breach
4	Payment-cadence irregularity detection	Ministry, State Nodal	Payment released disproportionate to physical progress reported
5	Collusion / repeated-vendor graph analysis	Ministry (investigation)	Shared bank accounts, repeated-winner contractor patterns across districts
6	Geotagged photo vs. progress-claim verification	District Authority	Claimed completion % not matching visual evidence

7	Completion-lag tracking	All roles	Fully paid but never marked complete — a known real gap in self-reported completion
8	Composite, explainable Risk Index (0–100)	All roles	Single defensible score per project, with SHAP-based “why flagged” explanation
9	Role-based dashboards (MP / District / State / Ministry)	All roles	Right-sized visibility per stakeholder, avoiding alert fatigue
10	RAG-based natural-language query assistant	All roles	“Why was this project flagged?” answered in plain language, grounded in MPLADS guidelines
11	Two-tier alert governance	Ministry, Platform Admin	Prevents false-positive-driven trust erosion
12	BHASHINI-powered regional-language UI	District/Implementing Agency field staff	Removes language barrier at data-entry point
13	Offline-first data capture + sync	Field-level Implementing Agency staff	Resilience against rural connectivity gaps
14	Platform self-monitoring (uptime/latency dashboard)	Platform Admin	Prevents the IT-portal-style trust collapse under load

4. Technical Architecture

4.1 Layer 1 — Unified Data Fabric

- **Ingestion:** Apache Kafka (partitioned by state/district) or Apache NiFi pulling from eSAKSHI MIS, PFMS payment records, GST/vendor registries, geotagged photo uploads, GIS coordinates
- **Storage:** Data lake (Delta Lake / Parquet on cloud object storage) for raw + curated layers; a **Knowledge Graph (Neo4j)** modeling relationships: MP → Constituency → District Authority → Implementing Agency → Contractor → Work → Payment — this

graph is what makes collusion and duplicate detection tractable, not just tabular anomaly scoring

- **Data quality gate:** schema validation + completeness checks at ingestion, since false positives downstream are disproportionately caused by upstream data quality issues (per §2.2)

4.2 Layer 2 — ML Detection Engine

Given the **2023–24-onward data constraint**, this is deliberately a hybrid, staged approach rather than a deep-learning-first pitch:

Model	Technique	Status Given Data Constraints
Duplicate/overlap detection	Sentence-BERT embeddings on work descriptions + DBSCAN geospatial clustering on lat/long	Deployable now — doesn't require years of history
Cost anomaly detection	Isolation Forest / statistical Z-score per work-category, region-adjusted	Deployable now — works on cross-sectional data
Cross-stage consistency checks	Rule-based + statistical: payment % vs. progress % vs. completion-flag mismatches	Deployable now, highest near-term ROI — exploits the multi-actor entry structure directly
Delay/overrun prediction	Prophet / gradient-boosted survival models initially; LSTM as data accumulates	Phased — statistical baseline now, deep sequence model once 4–5 years of history exist
Collusion detection	GraphSAGE (Graph Neural Network) over the contractor-agency-MP graph	Phased/roadmap — GNNs need denser relational history than currently available; pilot on synthetic + available data, scale as graph density grows
Photo-progress verification	CNN/Vision-Transformer comparing stage-wise geotagged photos to claimed progress %; optional satellite cross-check (Bhuvan/Sentinel-2) for large works	Deployable in pilot on a subset of high-value works first

4.3 Layer 3 — Composite Risk Scoring

- Weighted ensemble: rule-based guideline-compliance checks + ML anomaly scores → single 0–100 Risk Index

- **Explainability via SHAP/LIME** attached to every score — required both for investigator usability and legal defensibility (an unexplained AI accusation against an elected official is a non-starter)
- **Two-tier promotion logic** (per §2.2): Tier 2 (human-visible) alerts require corroboration across ≥ 2 independent signal types

4.4 Layer 4 — Role-Based Dashboards & Assistant

- Dashboarding: Apache Superset or Power BI Embedded, role-scoped views per §2.3
- RAG assistant: LangChain + vector DB (guideline documents, past investigation outcomes) for natural-language Q&A
- Regional-language layer: BHASHINI API integration for both data entry and dashboard text
- Meta-monitoring dashboard: Prometheus/Grafana tracking platform uptime, ingestion latency, alert precision-per-type

4.5 Tech Stack Summary

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Ingestion:      Kafka / NiFi
Storage:        Delta Lake (Parquet) + Neo4j (graph)
Processing:     Apache Spark
ML:             Isolation Forest, Sentence-BERT, DBSCAN, Prophet, GraphSAGE (phas
Explainability: SHAP / LIME
Dashboards:     Superset / Power BI
NL Assistant:   LangChain + vector DB (RAG)
Language:       BHASHINI API
Meta-monitoring: Prometheus + Grafana
Optional trust layer: Hyperledger (immutable audit log anchoring)

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5. Build Roadmap

Phase 0 — Data & Access Audit (Weeks 1–2)

- Formal data-sharing engagement with MoSPI/NIC for eSAKSHI/PFMS access or sandbox/synthetic data if direct access isn't available
- Schema mapping of actual portal fields (recommendation, sanction, progress, payment, completion, photo metadata) — **note: this requires live-rendering the JS-based dashboard or securing an API/export, since the public dashboard doesn't expose**

raw downloadable tables

- Data quality baseline audit (completeness, consistency) — this determines which Phase 1 models are viable on day one

Phase 1 — MVP: Rule-Based + Statistical Detection (Weeks 3–6)

- Cross-stage consistency checks (payment vs. progress vs. completion mismatches) — highest ROI, lowest data requirement
- Cost anomaly detection (Isolation Forest / Z-score)
- Duplicate/overlap detection (Sentence-BERT + geospatial clustering)
- Basic role-based dashboard (Ministry + one pilot state)
- Composite Risk Index v1 (rule-weighted, not yet ML-ensembled)

Phase 2 — ML Enhancement & Explainability (Weeks 7–10)

- SHAP explainability layer on all Tier 2 alerts
- Two-tier alert governance implementation
- Photo-progress verification pilot on high-value works subset
- Delay-prediction statistical baseline (Prophet)

Phase 3 — Advanced Detection & Trust Features (Weeks 11–16)

- Knowledge graph build-out (Neo4j) and GraphSAGE collusion-detection pilot
- RAG natural-language assistant
- BHASHINI regional-language integration
- Offline-first mobile data capture
- Meta-monitoring/self-observability dashboard

Phase 4 — Pilot & Feedback Loop (Weeks 17–20)

- Deploy to 1–2 pilot states/districts
- Human-review feedback loop wired into monthly retraining
- Alert-precision audit dashboard live
- Iterate based on District Authority and MP user feedback

6. Success Metrics

- % reduction in manual audit time per project
- Number/₹ value of duplicate works caught **pre-sanction** (highest-value intervention point)
- Days saved via delay prediction vs. reactive detection
- Alert precision rate (Tier 2 confirmed vs. dismissed) — tracked explicitly to avoid the false-positive trust erosion documented in §2.2
- Platform uptime/latency SLA adherence
- Regional-language adoption rate among District/Implementing Agency users

This corpus consolidates the problem statement analysis, live data-source review (eSAKSHI portal), external research on government-portal and fraud-detection failure patterns, the full feature set, technical architecture, and phased build plan as developed through this working session.